

Showcasing the BVAR Package with Global Volatility Spillovers

Final presentation inspired by Kuschnig & Vashold (2021)

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15 April 2026

Bayesian Econometrics | WU Vienna

Main Results

Why BVAR for This Project?

Econometric challenge

A VAR with M variables and p lags has

$$M + M^2 p$$

slope parameters before the covariance matrix.

With $M = 6$ implied-volatility indices and $p = 5$ lags, this already means

$$6 + 6^2 \cdot 5 = 186$$

coefficients to regularise.

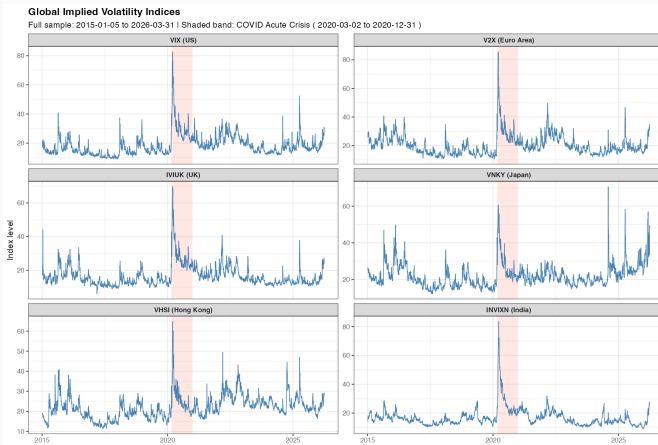
Why the package matters

- Hierarchical prior selection chooses shrinkage from the data.
- The same workflow estimates the model and produces IRFs, FEVDs, and scenario forecasts.
- This makes BVAR useful both for teaching and for applied work.

Our angle

We use global implied-volatility indices as a compact case study to show what the package can do in practice.

Data and Empirical Setting



Setup

- Six implied-volatility indices:
- VIX, V2X, IVIUK, VNKY, VHSI, INVIXN
- Full sample: 2015-01-05 to 2026-03-31
- Crisis window: 2020-03-02 to 2020-12-31

What the figure shows

- All indices spike together in early 2020.
- Levels differ persistently across regions.
- The common crisis spike makes cross-market spillovers economically plausible.

```
mn <- bv_minnesota(  
  lambda = bv_lambda(mode = 0.2, sd = 0.4),  
  alpha = bv_alpha(mode = 2),  
  psi = bv_psi(scale = 0.004, shape = 0.004)  
)  
  
fit <- bvar(  
  data = data_full, lags = 5,  
  n_draw = 25000, n_burn = 10000,  
  priors = bv_priors(hyper = c("lambda", "alpha", "psi"), mn = mn),  
  irf = bv_irf(horizon = 20, fevd = TRUE, identification = TRUE),  
  fcast = bv_fcast(horizon = 20)  
)  
  
irf_full <- irf(fit, horizon = 20, conf_bands = c(0.05, 0.16))  
fevd_full <- fevd(irf_full, conf_bands = c(0.16))
```

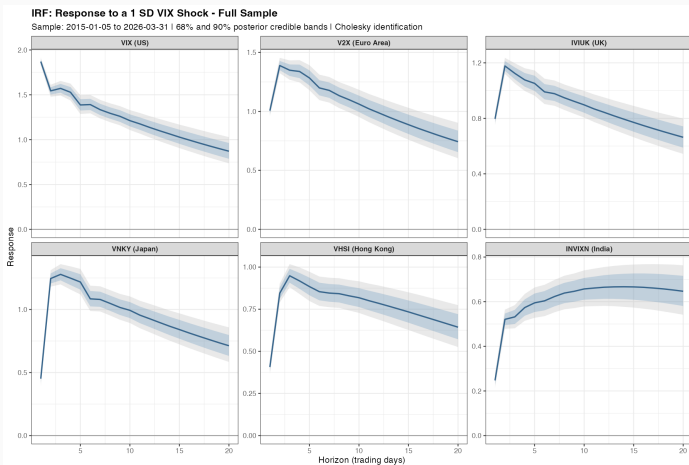
Why this is useful

- Prior setup, estimation, diagnostics, and post-estimation all live in one API.
- The package makes hierarchical shrinkage operational instead of purely theoretical.
- Conditional forecasts are easy to implement through `bv_fcast()`.

Outputs we focus on

- IRFs for dynamic transmission
- FEVD for variance attribution
- Scenario forecasts for stress testing

Full-Sample IRFs: 1 SD VIX Shock



What we learn

- Every market reacts positively to a VIX shock: spillovers are global, not local.
- Europe, the UK, and Japan respond strongly on impact.
- India reacts more gradually, but the response remains positive over the full horizon.
- The slow decay is consistent with persistent volatility dynamics.

Full-Sample FEVD



What we learn

- VIX is the dominant shock in most panels, especially for V2X and IVIUK.
- Japan and Hong Kong retain noticeable regional components, so transmission is strong but not exclusive.
- India remains much more self-driven than the other markets.
- FEVD complements the IRFs: the US shock is not only positive, but quantitatively important.

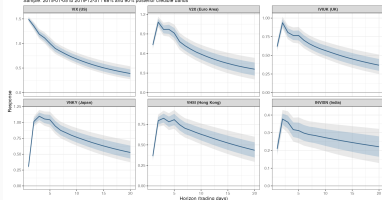
Crisis vs Calm: IRFs

IRF Comparison Across Regimes

Calm sample: 2015-01-05 to 2019-12-31 | Crisis sample: 2020-03-02 to 2020-12-31

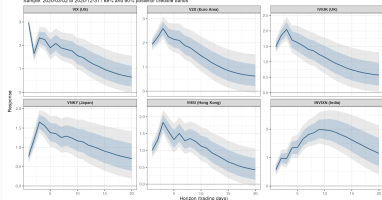
IRF: Response to a 1 SD VIX Shock - Calm Period

Sample: 2015-01-05 to 2019-12-31 | 68% and 90% posterior credible bands



IRF: Response to a 1 SD VIX Shock - COVID Acute Crisis

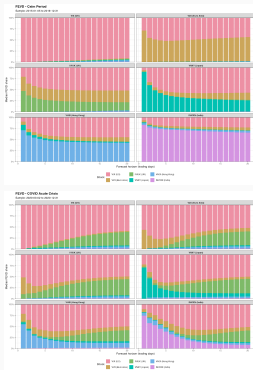
Sample: 2020-03-02 to 2020-12-31 | 68% and 90% posterior credible bands



What changes in crisis?

- Crisis responses are larger on impact and more uncertain.
- Europe and the UK react much more strongly during 2020-03-02 to 2020-12-31 than in 2015-01-05 to 2019-12-31.
- India becomes far more responsive and peaks later, suggesting delayed but substantial transmission.
- The sign stays positive in both regimes, but the magnitude depends strongly on market conditions.

Crisis vs Calm: FEVD



What changes in crisis?

- The crisis period shifts variance shares away from own shocks toward cross-market spillovers.
- VIX remains the single most important shock, but IVIUK gains importance in several panels during the crisis.
- Japan, Hong Kong, and India still keep regional components, yet these become relatively smaller in stressed markets.

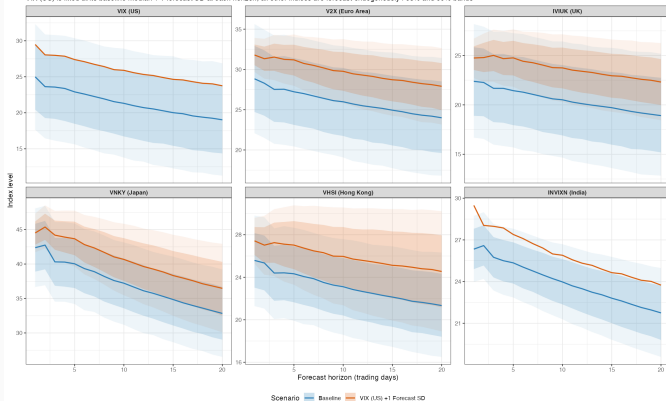
Interpretation

Contagion becomes broader in crisis periods: variance is explained less by purely domestic shocks and more by common external stress.

Conditional Forecast: Persistent VIX Stress

Baseline vs Conditional Forecast: VIX (US) Stress Scenario

VIX (US) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices are forecast endogenously | 68% and 90% bands



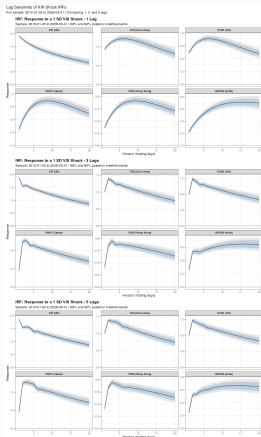
What we learn

- Fixing VIX one forecast standard deviation above baseline lifts every other forecast path.
- V2X, IVIUK, and VNKY show the largest upward shifts.
- The gap to baseline persists over the full forecast horizon, so this is not just a one-day response.

Why this showcases the package

The scenario is implemented directly with `bv_fcast()` and `predict()`, so stress testing is part of the package workflow.

Robustness to Lag Length



What we learn

- The ranking of responses is stable across 1, 3, and 5 lags.
- Shorter lag models smooth the early dynamics.
- Five lags allow sharper short-run spillovers, but the economic message stays the same.

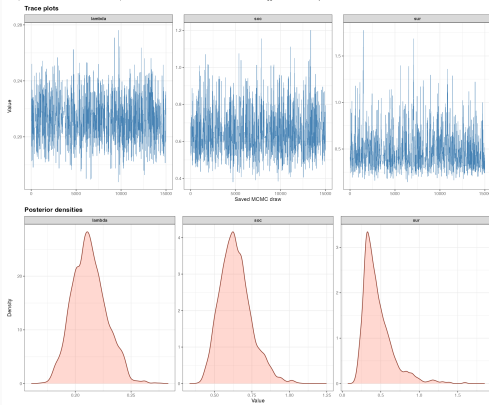
Bottom line

Our conclusions are not an artefact of one particular lag choice.

Adding SOC and SUR Leaves the Main Story Intact

Hyperparameter Trace and Density Diagnostics - Full Sample

Sample: 2010-01-01 to 2020-03-31 | Posterior prior: Minnesota + SOC + SUR | Geweke z outside +/- 1.96 suggests non-stationarity.



What changed?

In `BE_Group3_Code.R` we also estimated a Minnesota + SOC + SUR specification. The IRFs and FEVDs are visually almost unchanged, so the extra priors do not alter the economic story.

Why the change is small

SOC and SUR add level-VAR dummy observations:

$$y^+ = \text{diag}(\bar{y}/\mu), \quad x^+ = [0, y^+, \dots, y^+]$$

$$y^{++} = \bar{y}^\top / \delta, \quad x^{++} = [1/\delta, y^{++}, \dots, y^{++}]$$

For persistent volatility levels, these priors mostly reinforce dynamics already present in the data.

Convergence takeaway

The trace plots for λ , SOC, and SUR fluctuate around stable regions and the posterior densities are concentrated and roughly unimodal. This supports using Minnesota + SOC + SUR as a robustness check.

What This Shows About the BVAR Package

- BVAR turns a full Bayesian workflow into a compact set of functions: prior setup, estimation, diagnostics, IRFs, FEVDs, and scenario forecasts.
- In our application, the package reveals a clear empirical story: US volatility is the main transmitter, crisis periods amplify spillovers, and sustained VIX stress shifts all markets upward.
- Robustness checks do not overturn the conclusions: results are stable across lag choices and remain very similar when SOC and SUR are added.
- This is exactly why the package is useful in practice: it makes a theoretically rich Bayesian VAR framework fast to estimate, easy to diagnose, and easy to communicate.

Appendix: VAR Setup

Model

$$y_t = a_0 + A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma)$$

Why OLS is not enough

- Parameter count rises quickly with the number of variables and lags.
- Financial volatility series are persistent and strongly correlated.
- Unregularised VARs can overfit in-sample noise.

Why Bayesian shrinkage helps

- Priors shrink weak coefficients toward economically sensible benchmarks.
- The data can still dominate when the signal is strong.
- This improves stability and forecast performance in medium-size systems.

Appendix: Minnesota Prior

Prior mean

$$\mathbb{E}[(A_s)_{ij} \mid \Sigma] = \begin{cases} 1 & \text{if } i = j, s = 1 \\ 0 & \text{otherwise} \end{cases}$$

Prior variance

$$\text{Var}[(A_s)_{ij} \mid \Sigma] \propto \frac{\lambda^2 \psi_j}{s^{2\alpha}} \cdot \frac{\sigma_{ii}}{\sigma_{jj}}$$

- λ : overall tightness
- α : lag decay
- ψ_j : variable-specific scaling
- Intuition: own first lags are persistent; cross-lag effects are shrunk strongly toward zero

Appendix: Hierarchical Hyperparameter Selection

Key idea

Treat the prior hyperparameters as random variables:

$$p(\boldsymbol{\theta}, \boldsymbol{\gamma} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{\gamma}) p(\boldsymbol{\gamma}), \quad p(\boldsymbol{y} \mid \boldsymbol{\gamma}) = \int p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\gamma}) p(\boldsymbol{\theta} \mid \boldsymbol{\gamma}) d\boldsymbol{\theta}$$

Why this matters

- Shrinkage is learned from the data instead of fixed by hand.
- The marginal likelihood penalises overly flexible models automatically.
- Results become easier to justify and replicate.

How BVAR implements it

- Conjugate NIW structure gives closed-form posteriors for core parameters.
- Hyperparameters are sampled with Metropolis-Hastings.
- Diagnostics are available directly from the fitted object.

Appendix: SOC and SUR Dummy Priors

Sum-of-coefficients prior (SOC)

$$y^+ = \text{diag}\left(\frac{\bar{y}}{\mu}\right), \quad x^+ = [0, y^+, \dots, y^+]$$

Single-unit-root prior (SUR)

$$y^{++} = \frac{\bar{y}^\top}{\delta}, \quad x^{++} = \left[\frac{1}{\delta}, y^{++}, \dots, y^{++}\right]$$

- SOC shrinks the system toward “no large changes from current levels.”
- SUR allows persistent common movement and is useful when cointegration-like behaviour is plausible.
- These priors are designed for VARs in levels, which is why we only use them as a level-data robustness extension.

Appendix: Estimation Choices in Our Script

Baseline specification

- Six volatility indices in levels
- Lags: $p = 5$
- Draws: 25,000
- Burn-in: 10,000
- IRF / forecast horizon: 20 trading days

Identification and samples

- Recursive ordering:
- $VIX \rightarrow V2X \rightarrow IVIUK \rightarrow VNKY \rightarrow VHSL \rightarrow INVIXN$
- Calm sample: 2015-01-05 to 2019-12-31
- Crisis sample: 2020-03-02 to 2020-12-31

Interpretation note

All results should be read as reduced-form, recursively identified volatility transmission patterns rather than structural causal effects.

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